Week 8 Level 1: Introduction

Loops in Python

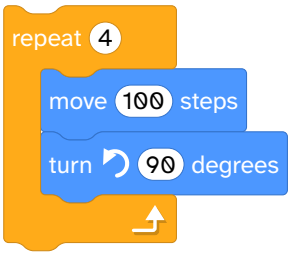
A `for` loop plus pushed-in lines is Scratch's repeat block, written in words.

What you will learn

- How to repeat lines with `for i in range(4):`.
- Why the spaces at the start of a line matter (indentation).
- The secret rule for drawing any shape with equal sides.
- How to put a loop inside a loop to make patterns.

The big idea: repeat, in words

Last week your square needed 8 lines. With a loop it needs 3. Compare:



→

```
for i in range(4):
    t.forward(100)
    t.left(90)
```

Python has no C-shaped block. Instead it uses two clues:

1. a **colon** `:` at the end of the `for` line, and
2. **4 spaces** at the start of every line **inside** the loop. This is called **indentation**.

```
for i in range(3):
    t.forward(100)
    t.left(120)
print("Done!")
```

4 spaces

inside the loop: repeats 3 times

after the loop: runs once

Tip: Thonny helps you

After a colon, press `Return`: Thonny pushes the next line in for you. To come back out of the loop, press `Delete` once.

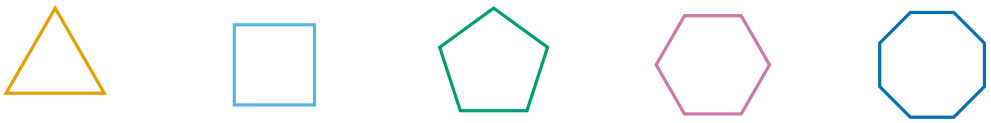
How many times?

`range(4)` gives Python four numbers: `0`, `1`, `2`, `3`. Python starts counting at `0`! What matters is that there are **4** of them, so the loop runs 4 times.

The secret shape rule

To draw a shape and get back to the start, the turtle turns all the way round: 360 degrees. Share that turn between the corners:

turn at each corner = 360 ÷ number of sides



triangle	square	pentagon	hexagon	octagon
3 sides	4 sides	5 sides	6 sides	8 sides
turn 120	turn 90	turn ?	turn ?	turn ?

Fill in the missing turns: pentagon _____, hexagon _____, octagon _____.

Exercise 8.1: Loop Square

Folder: `week08/ex-loop-square`. Run the starter: the turtle walks in a straight line! Add `t.left(90)` **inside** the loop, lined up under `t.forward(100)`. Run it, then open `check_my_work.py` and press Run.

Exercise 8.2: Polygon Party

Folder: `week08/ex-polygon-party`. The triangle is done. Write a loop for a **square** (sides of 100) and a loop for a **hexagon** (sides of 60). Use the shape rule for the turns. Done looks like: an orange triangle, a sky-blue square and a green hexagon, all closed.

A loop inside a loop

A loop can hold another loop. The **outer** loop picks a color. The **inner** loop draws a whole square. Then the turtle turns a little and the outer loop goes again.

```
for color in PALETTE:          # 8 colors: runs 8 times
    t.color(color)
    for side in range(4):       # draws one whole square
        t.forward(100)
        t.left(90)
    t.left(45)                  # 8 x 45 = 360
```

The inner lines have **8** spaces: 4 for the outer loop and 4 more for the inner loop. How many lines does the turtle draw altogether? _____

Exercise 8.3: Square Flower

Folder: `week08/ex-square-flower`. The starter draws one square. Put the square loop inside a color loop (like the code above) to make 8 squares in 8 colors. Tip: `t.speed(0)` draws super fast.

Tip: Asking for help well

Stuck? First try "three before me": read the red message, check this workbook, ask your partner. Then ask a grown-up with the help sentence:
I expected _____, but _____ happened. I tried _____.

Vocabulary box

Loop Code that repeats.

for The Python word that starts a loop: `for i in range(4):`.

range Gives the loop its numbers. `range(4)` means 0, 1, 2, 3: four times.

Indentation The spaces at the start of a line. They show which lines are inside the loop.

Nested loop A loop inside another loop.

Off-by-one bug When a loop runs one time too many or one time too few.

Reflection

How is `for i in range(4):` like Scratch's repeat block? How is it different?

Draw (or describe) the coolest pattern you made today:

Week 8 Level 1: Introduction

Loop Fixer

15–20 minutes. Do Core first; Challenge and Creative are optional.

Name (nickname is fine): _____ Date: _____

★ Core (everyone): Loop Fixer (about 10 minutes)

Open `week08/hw-loop-fixer/starter.py`. It should draw a blue square and an orange star, but it has **2 loop bugs**.

1. Press Run and watch carefully. Count the sides of the square. How many times does its loop repeat?
2. The star is a straight line! Look at the spaces at the start of each line. Which line is **not** inside the loop?
3. Fix one bug, press Run, then fix the other. Finish with `check_my_work.py`.

Bug 1 was on line _____. I changed _____ to _____.

Bug 2 was on line _____. I fixed it by _____.

No computer at home? Circle the two bugs in the program below and write the fixes next to them.

```
for i in range(3):
    t.forward(100)
    t.left(90)

for i in range(5):
    t.forward(150)
t.right(144)
```

★★ Challenge (want more?): Count the taps

Without running it: how many times does this program print `tap`? _____

```
for i in range(3):
    for j in range(2):
        print("tap")
```

Then make a 7-pointed star: 7 lines of 150, turning `t.right(360 * 3 / 7)` each time (about 154 degrees).

Creative (make it yours): Spiral lab

Make a new file `my-python/spiral-lab.py` with this loop (plus `import turtle`, `t = turtle.Turtle()` and `turtle.done()`). Change the turn number and record what you see.

```
for i in range(60):
    t.forward(i * 3)
    t.left(91)
```

Turn	What it looked like
91	
121	
my choice:	

Self-check: tick each box when it is true

- ☐ My square has 4 sides and my star has 5 points.
- ☐ I can explain why the spaces at the start of a line matter.
- ☐ I know that `range(4)` makes a loop run 4 times.
- ☐ If I got stuck, I used the help sentence: “I expected … but … happened.”